HIGH-SPEED APTITUDE DRILL

Formulaic Shortcuts & Verified Answer Key (Sets 1-4)

SET 1 SOLUTIONS

Section A: Quantitative Aptitude

Q1. SP at 15% loss = CP × 0.85 = ■1,200 × 0.85 = **■1,020**.	**A) ■1,020**
Q2. Combined Work Time = (A × B) / (A + B) = (12 × 18) / (12 + 18) = 216 / 30 = **7.2 days**.	**C) 7.2 days**
Q3. SI = (P × R × T) / 100 = (6000 × 5 × 4) / 100 = **■1,200**.	**B) ■1,200**
Q4. Removed Number = (Old N × Old Avg) - (New N × New Avg) = (5 × 24) - (4 × 22) = 120 - 88 = **32**.	**B) 32**
Q5. Ratio units sum = 3 + 5 = 8. 1 unit = 160 / 8 = 20. Larger number = 5 × 20 = **100**.	**D) 100**
Q6. Speed in m/s = 54 × (5/18) = 15 m/s. Time = Distance / Speed = 150 / 15 = **10 seconds**.	**A) 10 sec**
Q7. Original Number = Result / 1.20 = 180 / 1.2 = **150**.	**B) 150**
Q8. LEVEL has 5 letters (2 L's, 2 E's, 1 V). Total arrangements = 5! / (2! × 2!) = 120 / 4 = **30**.	**B) 30**
Q9. Downstream Speed = Speed of boat + Stream = 12 + 3 = 15 km/h. Time = 45 / 15 = **3 hours**.	**C) 3 hours**
Q10. LCM(12, 18, 24): Multiples of 24 are 24, 48, 72. 72 is perfectly divisible by both 12 and 18. LCM = **72**.	**C) 72**

Section B: Logical Reasoning

Q11. Series grows by consecutive even integers (+4, +6, +8, +10, +12). Next term = 30 + 12 = **42**.	**C) 42**
Q12. Shift forward pattern (+1 letter). C→D, A→B, T→U. Following same for DOG: D→E, O→P, G→H ⇒ **EPH**.	**A) EPH**
Q13. Mother's only brother = Riya's maternal uncle. His son is Riya's maternal **cousin**.	**C) Cousin**
Q14. Opposite 5 km north and 5 km south movements perfectly cancel out. Net displacement is the eastward run = **3 km**.	**B) 3 km**
Q15. Standard Syllogism. 'Some books are pencils' converts directly by equivalence to **Some pencils are books** (Conclusion II). Conclusion I cannot be assumed.	**B) Only II**
Q16. All terms are perfect cubes (2^3, 3^3, 4^3, 5^3) except 100, which is a perfect square (10^2).	**D) 100**
Q17. Total = (Top Rank + Bottom Rank) - 1. Bottom Rank = 40 - 12 + 1 = **29**.	**C) 29**
Q18. Formula $\theta = \|30H - 5.5M\| = \|30(3) - 5.5(30)\| = \|90 - 165\| =$ **75°**.	**B) 75°**
Q19. Days gap = 1st Feb - 5th Jan = 27 days. Remainder of 27 / 7 = 6 odd days. Monday + 6 days = **Sunday**.	**C) Sunday**
Q20. Alphabet gaps expand progressively: A (+2) → C (+3) → F (+4) → J (+5) → O (+6) → **U**.	**B) U**

SET 2 SOLUTIONS

Section A: Quantitative Aptitude

Q1. CI = P[(1 + R/100)² - 1] = 10,000[(1.1)² - 1] = 10,000 × 0.21 = ■**2,100**.	**B) ■2,100**
Q2. Combined Time = (A × B) / (A + B) = (10 × 15) / (10 + 15) = 150 / 25 = **6 hours**.	**D) 6 hours**
Q3. Alligation method: (Cheaper vs Dearer) Ratio = (30 - 24) : (24 - 20) = 6 : 4 = **3 : 2**.	**B) 3 : 2**
Q4. Let son = x, Father = 3x. In 12 years: 3x + 12 = 2(x + 12) ⇒ 3x + 12 = 2x + 24 ⇒ x = **12 years**.	**B) 12 years**
Q5. Average Speed (Equal Distances) = 2xy / (x + y) = (2 × 40 × 60) / (40 + 60) = 4800 / 100 = **48 km/h**.	**C) 48 km/h**
Q6. Pairs yielding sum 7: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) = 6 outcomes. Probability = 6 / 36 = **1/6**.	**D) 1/6**
Q7. Combinations formula ■C■ = (8 × 7 × 6) / (3 × 2 × 1) = **56**.	**B) 56**
Q9. Net change for successive equal % change x = $-x^2 / 100 = -(20)^2 / 100$ = **4% decrease**.	**D) 4% dec**
Q10. Profit split matches investment ratio 4,000 : 6,000 = 2 : 3. A's share = (2/5) × ■5,000 = ■**2,000**.	**A) ■2,000**

Section B: Logical Reasoning

Q11. Pattern: (Current × 2) + 1. Next value = (47 × 2) + 1 = 94 + 1 = **95**.	**C) 95**
Q12. Analogy matching workplace context. A doctor works inside a hospital; a teacher works inside a **school**.	**B) School**
Q13. Seating layout alignment from left constraint: [Q, R, P, T, S]. Middle slot is occupied by **P**.	**D) P**
Q14. Summing alphabetical positions: D(4) + E(5) + A(1) + F(6) = **16**.	**A) 16**
Q15. Total Union n(T∪C) = 60 + 50 - 20 = 90. Neither = Total - Union = 100 - 90 = **10**.	**B) 10**
Q16. Pattern: Row(Col■)² + Row(Col■)² = Row(Col■). Row 3 calculation: $4^2 + 5^2 = 16 + 25 = 41$.	**A) 41**
Q17. Assumption I directly supports the core objective of policy promotion. Assumption II is an absolute statement that isn't logical.	**A) Only I**
Q18. Letter blocks shift by ignoring 1 character: AB (skip C) DE (skip F) GH (skip I) JK (skip L) **MN**.	**B) MN**
Q19. Cubes with exactly two faces painted lie exclusively along the edges: 12 × (s - 2) = 12 × (3 - 2) = **12**.	**B) 12**
Q20. Applying rules: 8 × 3 ÷ 6 + 4 - 2. Solve Bodmas: (24 ÷ 6) + 4 - 2 = 4 + 4 - 2 = **6**.	**D) 6**

SET 3 SOLUTIONS

Section A: Quantitative Aptitude

Q1. Net Multiplier = 0.80 × 0.90 = 0.72. Final SP = ■500 × 0.72 = **■360**.	**C) ■360**
Q2. Calculation: (0.25 × 80) + (0.40 × 50) = 20 + 20 = **40**.	**A) 40**
Q3. Using cyclicity or mod math: $2^3 \equiv 1 \pmod 7$. Therefore, 2^{1}■ = $(2^3)^3 \times 2^1 \equiv 1^3 \times 2 = 2$.	**B) 2**
Q4. Volume = $\pi r^2 h$ = (22/7) × 7 × 7 × 10 = 22 × 7 × 10 = **1,540 cm³**.	**D) 1,540**
Q5. Numbers are x = (40+10)/2 = 25 and y = (40-10)/2 = 15. Product = 25 × 15 = **375**.	**C) 375**
Q7. Relative speed = 60 + 40 = 100 km/h = 100 × (5/18) = 250/9 m/s. Crossing Time = Total Length / Rel Speed = (120 + 180) / (250/9) = 300 × 9 / 250 = **10.8 seconds**.	**D) 10.8 sec**
Q8. Sort items: [3, 5, 7, 9, 12, 15]. Median (even count mid-average) = (7 + 9) / 2 = **8**.	**C) 8**
Q9. AP formula S_n = n/2 * [2a + (n-1)d]. Here n=10, a=3, d=4. S■■ = 5 × [6 + (9×4)] = 5 × 42 = **210**.	**A) 210**

Section B: Logical Reasoning

Q11. Alternating series structures: Series 1: 3, 6, 9 (+3 rule). Series 2: 8, 11, 14 (+3 rule). Next is from Series 1: 9 + 3 = **12**.	**B) 12**
Q12. Reversed pair alphabet map (A↔Z, B↔Y). CHAIR converts letter-for-letter to **XSZRI**.	**C) XSZRI**
Q13. Arranging 6 elements sequentially along a circle puts index elements i and i+3 completely opposite. C pairs directly across from **F**.	**A) F**
Q14. From P > Q ≥ R = S, we trace P > S (Conclusion I fits) and Q ≥ S (Conclusion II fits). Both hold true.	**D) Both follow**
Q15. Standard grid square counting matrix: $1^2 + 2^2 + 3^2 = 1 + 4 + 9 =$ **14 total squares**.	**B) 14**
Q17. Chronological lifecycle path: Seed (1) → Plant (2) → Tree (5) → Flower (4) → Fruit (3) ⇒ **1 2 5 4 3**.	**B) 1 2 5 4 3**
Q18. Actual Time mapping formula = 11:60 - Mirror Time = 11:60 - 3:15 = **8:45**.	**B) 8:45**
Q19. Pattern math rule: n × (n + 1). 7 × 8 = 56. Following pattern for 9: 9 × 10 = **90**.	**D) 90**
Q20. Height comparisons: D > A > B > C. The shortest element in the chain is cleanly **C**.	**C) C**

SET 4 SOLUTIONS

Section A: Quantitative Aptitude

Q1. Total Distance = Speed × Time = 50 km/h × 6 hours = **300 km**.	**300 km**

This reference is optimized for high-speed calculation reviews. Cross-verified via algorithmic shortcuts.